“Calculus 3”

Multi-Variable Calculus

Instructor: Álvaro Lozano-Robledo

Day 24

Any Reminders? Any Questions?

- Quiz on Friday (discussion) covers 16.5, 16.6 (curl, divergence, parametric surfaces)

- Office hours Mondays 1-2pm online
- Office hours Thursday 3:30-4:30 at MONT 233
- Calc Night Thursday 6:30-8:30 at MONT 104

ALVARO: Start the recording!

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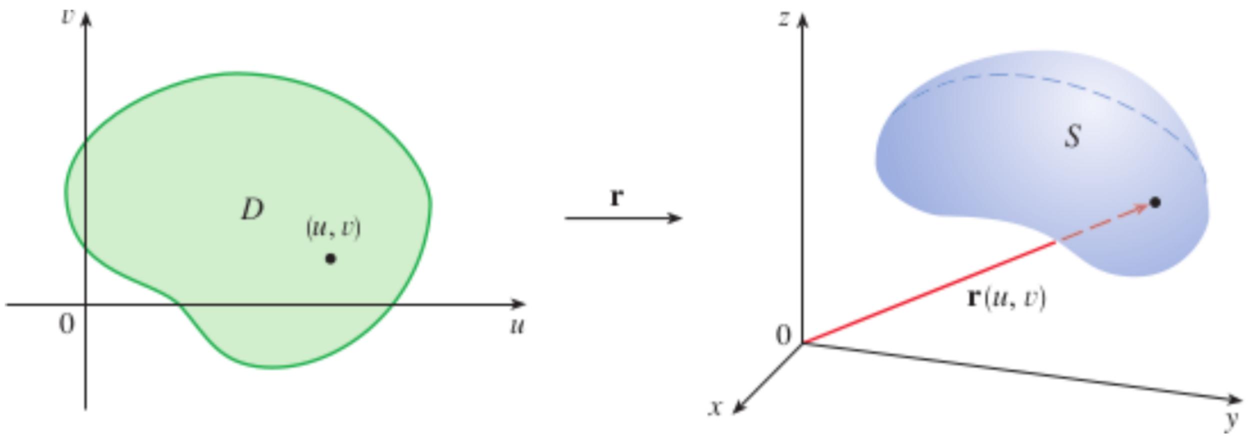
Surface Area

About Parametrized Surfaces and their Area

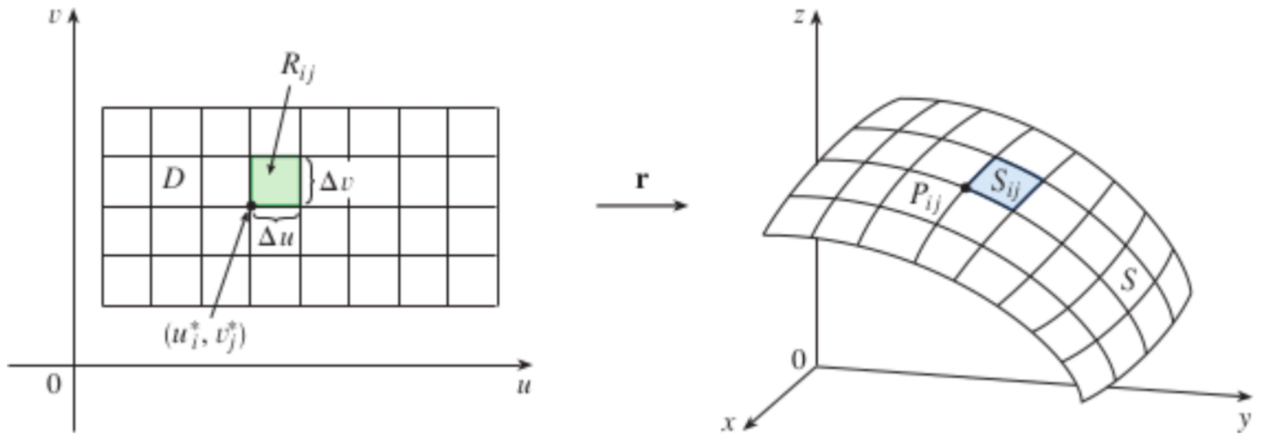
$$\mathbf{r}(u, v) = x(u, v)\,\mathbf{i} + y(u, v)\,\mathbf{j} + z(u, v)\,\mathbf{k}$$

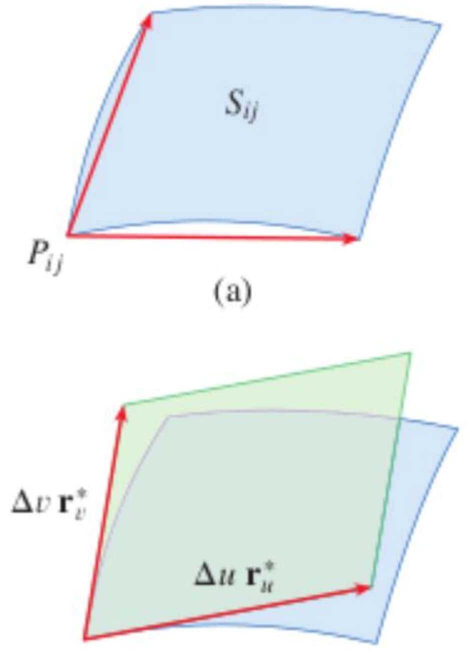
A parametric surface



About Parametrized Surfaces and their Area

The image of the subrectangle R_{ij} is the patch S_{ij}.





$$\left|(\Delta u\, \mathbf{r}_u^*) \times (\Delta v\, \mathbf{r}_v^*)\right| = \left|\mathbf{r}_u^* \times \mathbf{r}_v^*\right| \Delta u\, \Delta v$$

$$\sum_{i=1}^{m} \sum_{j=1}^{n} \left|\mathbf{r}_u^* \times \mathbf{r}_v^*\right| \Delta u\, \Delta v$$

About Parametrized Surfaces and their Area

If a smooth parametric surface S is given by the equation

$$\mathbf{r}\left(u,v\right)=x\left(u,v\right)\mathbf{i}+y\left(u,v\right)\mathbf{j}+z\left(u,v\right)\mathbf{k}\qquad\left(u,v\right)\in D$$

and S is covered just once as (u, v) ranges throughout the parameter domain D, then the **surface area** of S is

$$A\left(S\right)=\iint\limits_{D}\left|\mathbf{r}_u\times\mathbf{r}_v\right|dA$$

where

$$\mathbf{r}_u=\frac{\partial x}{\partial u}\mathbf{i}+\frac{\partial y}{\partial u}\mathbf{j}+\frac{\partial z}{\partial u}\mathbf{k}\qquad\mathbf{r}_v=\frac{\partial x}{\partial v}\mathbf{i}+\frac{\partial y}{\partial v}\mathbf{j}+\frac{\partial z}{\partial v}\mathbf{k}$$

Surfaces Area of a Graph (and Length of a Graph)

$$A(S) = \iint_D \sqrt{1 + \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2}\, dA$$

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$$

Example: Compute the surface area of the paraboloid $z = x^2 + y^2$ below the plane z = 9.

[Extra Space]

Example: Compute the surface area of the paraboloid $z = x^2 + y^2$ below the plane z = 9.

Surfaces Area of a Surface of Revolution

Surfaces Area of a Surface of Revolution

$$A = \iint\limits_{D} \left| \mathbf{r}_x \times \mathbf{r}_\theta \right| \, dA$$

$$= \int_0^{2\pi} \int_a^b f(x) \sqrt{1 + [f'(x)]^2} \, dx \, d\theta$$

$$= 2\pi \int_a^b f(x) \sqrt{1 + [f'(x)]^2} \, dx$$

Example: Compute the surface area of the paraboloid $z = x^2 + y^2$ below the plane z = 9, as a surface of revolution.

[Extra Space]

Example: Compute the surface area of the paraboloid $z = x^2 + y^2$ below the plane z = 9, as a surface of revolution.

Questions?

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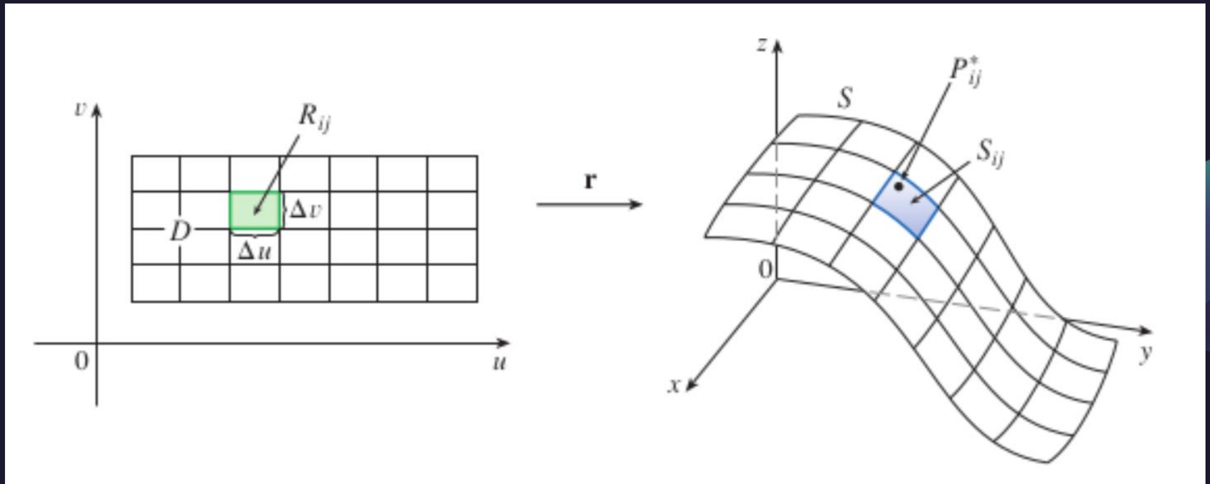
Surface Integrals

Integrals on Parametrized Surfaces

$$\mathbf{r}(u, v) = x(u, v)\,\mathbf{i} + y(u, v)\,\mathbf{j} + z(u, v)\,\mathbf{k}$$



$$\iint_S f(x, y, z)\; dS = \lim_{m,\, n \to \infty} \sum_{i=1}^{m} \sum_{j=1}^{n} f\left(P_{ij}^*\right)\; \Delta S_{ij}$$

$$\iint_S f(x, y, z)\, dS = \iint_D f(\mathbf{r}(u, v))\, |\mathbf{r}_u \times \mathbf{r}_v|\, dA$$

Example: Compute the surface integral $\iint_S x^2 \, dS$ where S is the unit sphere.

$$x = a \sin\phi \cos\theta \qquad y = a \sin\phi \sin\theta \qquad z = a \cos\phi$$

[Extra Space]

Example: Compute the surface integral $\iint_S x^2 \, dS$ where S is the unit sphere.

$x = a \sin\phi \cos\theta \qquad y = a \sin\phi \sin\theta \qquad z = a \cos\phi$

[Extra Space]

Example: Compute the surface integral $\iint_S x^2\, dS$ where S is the unit sphere.

$$\iint_S x^2\ dS = \iint_D (\sin\phi\,\cos\theta)^2\ |\mathbf{r}_\phi \times \mathbf{r}_\theta|\ dA$$

$$= \int_0^{2\pi}\int_0^{\pi} \sin^2\phi\,\cos^2\theta\,\sin\phi\ d\phi\ d\theta = \int_0^{2\pi}\cos^2\theta\ d\theta \int_0^{\pi}\sin^3\phi\ d\phi$$

$$= \int_0^{2\pi}\frac{1}{2}(1+\cos\,2\theta)\ d\theta \int_0^{\pi}(\sin\phi - \sin\phi\,\cos^2\phi)\ d\phi$$

$$= \frac{1}{2}\left[\theta + \frac{1}{2}\sin 2\theta\right]_0^{2\pi}\left[-\cos\phi + \frac{1}{3}\cos^3\phi\right]_0^{\pi} = \frac{4\pi}{3}$$

Integrals on Graphs of Functions

$$\mathbf{r}(u,v) = x(u,v)\,\mathbf{i} + y(u,v)\,\mathbf{j} + z(u,v)\,\mathbf{k}$$

$$\iint\limits_S f(x,y,z)\,dS = \iint\limits_D f(\mathbf{r}(u,v))\,\left|\mathbf{r}_u \times \mathbf{r}_v\right|\,dA$$

$$\iint\limits_S f(x,y,z)\,dS = \iint\limits_D f(x,y,g(x,y))\sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1}\,dA$$

Example: Compute the surface integral $\iint_S \; y \, dS$ where S is the surface

$$z = x + y^2 \, , \; 0 \le x \le 1, \;\; 0 \le y \le 2.$$

[Extra Space]

Example: Compute the surface integral $\iint_S \; y \, dS$ where S is the surface $z = x + y^2 \,,\; 0 \le x \le 1, \;\; 0 \le y \le 2.$

Example: Compute the surface integral $\iint_S z\, dS$ where S is the surface whose sides are given by the cylinder and functions:

$$1 = x^2 + y^2\ ,\ z = 0,\ \ z = 1 + x.$$

[Extra Space]

Example: Compute the surface integral $\iint_S \; z\, dS$ where S is the surface whose sides are given by the cylinder and functions:

$$1 = x^2 + y^2\,,\;\; z = 0,\;\; z = 1 + x.$$

Orientation

- For surface integrals of a vector field $\vec{F}$ over a surface S, the orientation of the surface must be specified, just like the line integrals. The orientation of a surface is specified via its normal vectors at any point.
- For a closed surface S, we define **the positive orientation** of S to mean that all normal vectors to S point outward (with negative meaning normal vectors point inward instead). If it is impossible to have all normal vectors always point either outward or inward, then the surface S is non-orientable.

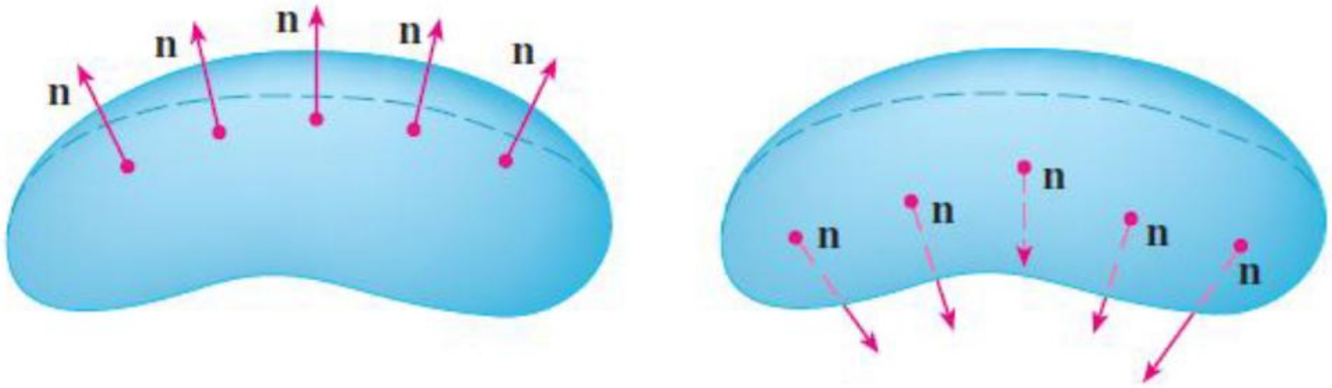
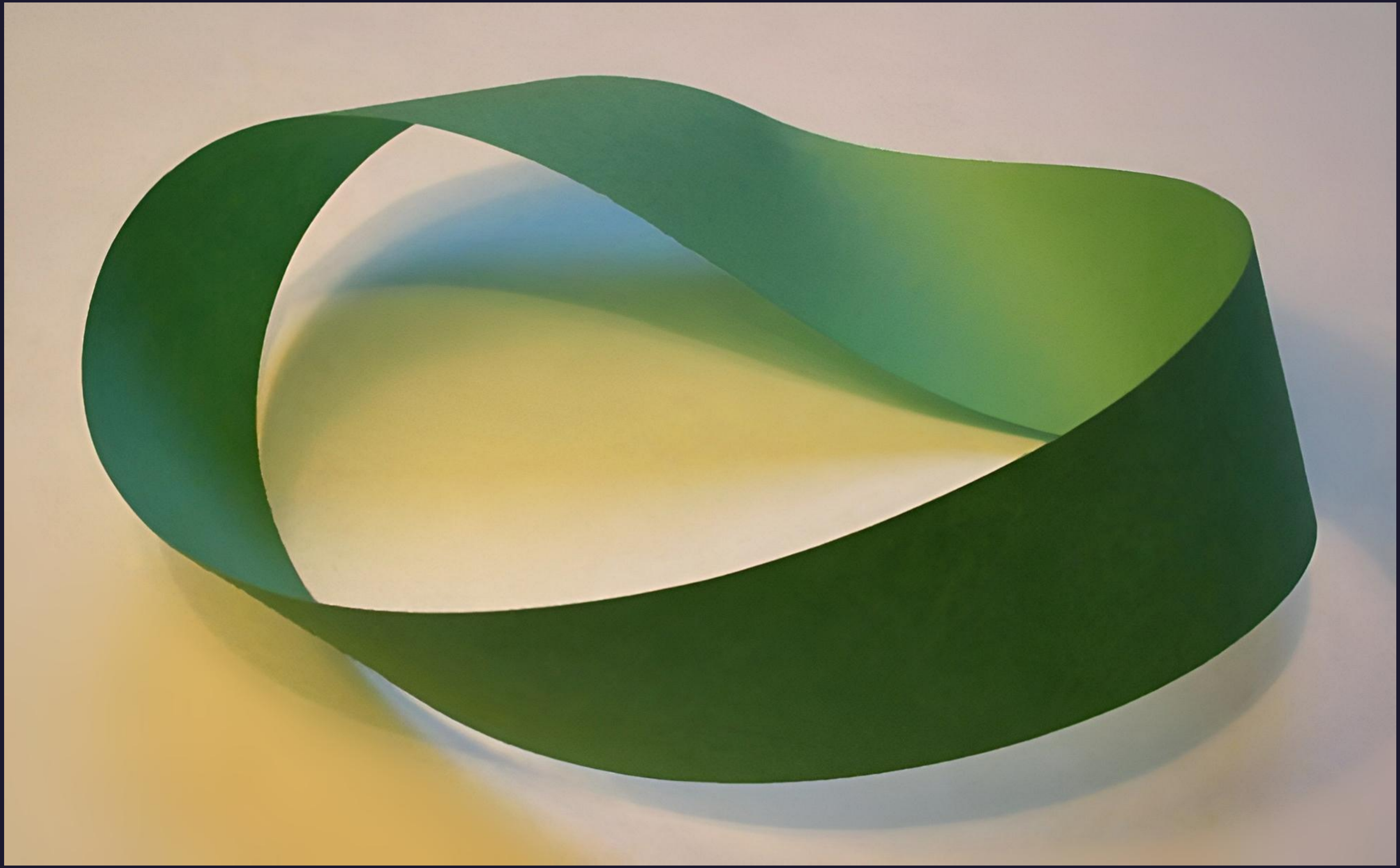


FIGURE 7
The two orientations of an orientable surface

Orientation



Orientation

- To define a surface integral of a vector field, we only want to count the portion of $\vec{F}$ that is normal to the surface S at any point. Hence, the dot product again plays a role in the computation.
- Then the **surface integral** of $\vec{F}$ over the surface S is defined as

$$\begin{aligned}\iint_S \vec{F}\cdot d\vec{S} &= \iint_S \vec{F}\cdot\vec{n}\,dS\\ &= \iint_S \vec{F}\cdot\frac{\vec{r}_u\times\vec{r}_v}{|\vec{r}_u\times\vec{r}_v|}dS\\ &= \iint_D [\vec{F}\cdot\frac{\vec{r}_u\times\vec{r}_v}{|\vec{r}_u\times\vec{r}_v|}]|\vec{r}_u\times\vec{r}_v|dA\\ &= \iint_D \vec{F}(\vec{r}(u,v))\cdot(\vec{r}_u\times\vec{r}_v)dA.\end{aligned}$$

This is often called the **flux integral** as it measures the flux of $\vec{F}$ across the surface S.

- Recall and compare it with the line integral of vector fields:

$$\int_C \vec{F}\cdot d\vec{r} = \int_a^b \vec{F}(\vec{r}(t))\cdot\vec{r}\,'(t)dt.$$

Example: Find the flux of the vector field $F = (0, y, -z)$ across the surface S given by $y = x^2 + z^2\,,\ 0 \leq y \leq 1.$

(Assume S is positively oriented.)

[Extra Space]

Example: Find the flux of the vector field $F = (0, y, -z)$ across the surface S given by $y = x^2 + z^2\,,\ 0 \leq y \leq 1.$

(Assume S is positively oriented.)

Example: Let F be the vector field $F = (z^2, x^2, y^2)$, and the surface S be given by $z = x^2 + y^2$, $1 \leq z \leq 9$, $y \geq 0$. Assume S is positively oriented. Set up (but do not evaluate).

$$\iint_S curl(F) \cdot dS$$

[Extra Space]

Example: Let F be the vector field $F = (z^2, x^2, y^2)$, and the surface S be given by $z = x^2 + y^2$, $1 \leq z \leq 9$, $y \geq 0$. Assume S is positively oriented. Set up (but do not evaluate).

$$\iint_S curl(F) \cdot dS$$

Questions?

Thank you

Until next time.